

Starting Minikube

Run the following command from a terminal:

```
$minikube start
```

In no time, the cluster will be fired up and running.

To check if Kubectrl and Minikube are installed properly, run the commands as shown in Figure 6.

```
$ kubectl get pods --all-namespaces
NAMESPACE   NAME                                     READY   STATUS    RESTARTS   AGE
kube-system  kube-addon-manager-minikube            1/1     Running   2           15h
kube-system  kube-dns-56ccf8bf8-5cnvv               3/3     Running   6           15h
kube-system  kubernetes-dashboard-770898585-xbzqp   1/1     Running   3           15h
kube-system  storage-provisioner                    1/1     Running   2           15h
kube-system  tiller-deploy-865d6c794-xgkzp          1/1     Running   2           15h
```

Figure 6: Get the list of the running pods

The output is a table with six columns. The first column indicates the namespace to which a pod belongs. The second column indicates the name of the pod. The third column shows the number of pods to be scheduled and the number of currently running pods. The fourth column shows the status of the pod. The fifth column shows the number of times the given pod has restarted. The last column shows the age of the pod, i.e., the time since the creation of the pod.

Now we have a Kubernetes cluster running locally. As you can see, we didn't have to configure the network or any other resource specifically. Minikube takes care of all the internal configurations. Hence, Minikube saves a lot of time when deploying applications in a development environment locally.

Stopping a cluster

Sometimes we may have to stop the cluster to free computing resources. In such a scenario, run the following command:

```
$minikube stop
```

Now that we have set up a cluster, let us investigate Helm, with which we can deploy applications like WordPress or MySQL on the cluster, with just a single command.

Using Helm

Helm is a package manager for Kubernetes that streamlines the process of installing and managing applications on the cluster. You can relate to it as *apt/ebuild/yum/homebrew* for Kubernetes. The name 'Helm' refers to the steering wheel of a ship. In the context of Kubernetes, instead of steering a ship, Helm steers the cluster, hence managing applications as illustrated in Figure 7.

Helm applications are written as Helm charts, which help us define, install and upgrade any Kubernetes application. Charts are generally preconfigured Kubernetes resources.

Helm can be used for the following tasks:

- Manage releases of applications
- Share applications as charts
- Manage Kubernetes manifest files



Figure 7: Illustrating how Helm is used (Logo sources: Respective websites)

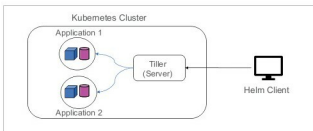


Figure 8: Helm architecture

- Search and use already existing applications packaged as charts

Helm comprises two parts—Helm (a client) and Tiller (a server), as shown in Figure 8. Tiller runs inside Kubernetes and manages the installation of Kubernetes charts. Helm can be installed on laptops, servers, or on continuous integration or continuous delivery tools like Jenkins or Chef.

Kubernetes charts can be stored on a disk or they can be fetched from remote repositories. A chart consists of two important things: *Chart.yml* (description of the chart) and one or more templates containing Kubernetes manifest files.

To use Helm in a secure and efficient way, you need a Kubernetes cluster, Helm and Tiller configuration. You need to decide what security configurations to apply to your installation such as controlling the cluster in network environments, multi-tenant clusters, clusters with access to important data, etc. More information on securing Helm can be found at https://docs.helm.sh/using_helm/#securing-your-helm-installation/.

Deploying applications using Helm

First, we need to install Helm in order to be able to deploy applications with it. To install Helm, run the command shown in Figure 9.

These commands will download the Helm script and save it as the *get_helm.sh* file, then the required permissions are given, and the script installs Helm.